# Cristian Castiglione

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## Research interests

Statistical computing, Mixed and Additive Models, Spatial Statistics, Matrix factorization, Network data.

## Current position

**Postdoctoral research fellow** — Milan, Italy
Bocconi University, Institute for Data Science and Analytics (BIDSA) — Apr 2025 – Mar 2027
Project: *sociogeNEsis of criMinal nEtworks: reconStruction, dIscovery and diSruption (NEMESIS)*
Grant: ERC Starting Grant
Advisor: Prof. Daniele Durante

## Past academic positions

**Postdoctoral research fellow** — Milan, Italy
Bocconi University, Institute for Data Science and Analytics (BIDSA) — Apr 2024 – Mar 2025
Project: *Causes of deAth dependence stRuctures and the cOmpositioNal effecT on ovErall mortality (CARONTE)*
Grant: PRIN–MUR 2022 Grant
Advisor: Prof. Daniele Durante

**Postdoctoral research fellow** — Padua, Italy
University of Padua, Department of Statistical Sciences — Feb 2023 – Apr 2024
Project: *Statistical methods and models for the integration of multiomic data*
Grant: PNRR Grant
Advisor: Prof. Davide Risso

## Education

**Ph.D.** **University of Padua**, Department of Statistical Sciences — Padua, Italy
Program: Statistical Sciences — Oct 2019 – May 2023
Thesis: *Approximate inference for misspecified additive and mixed models*
Advisors: Prof. Mauro Bernardi
Co-advisors: Prof. Laura M. Sangalli, Prof. Alessio Farcomeni

**M.S.** **University of Padua**, Department of Statistical Sciences — Padua, Italy
Course: Statistical Sciences — Oct 2016 – Nov 2018
Thesis: *Dynamic quantile models for spatio-temporal data*
Advisor: Prof. Mauro Bernardi
Final mark: 110/110 cum Laude

**B.S.** **University of Padua**, Department of Statistical Sciences — Padua, Italy
Course: Statistics, Economics and Finance — Oct 2013 – Jul 2016
Thesis: *Multistate models for competing risks*
Advisor: Prof. Giuliana Cortese
Final mark: 110/110

## Work experience

**Blue BI S.R.L.**, Junior consultant in business intelligence and analytics — Vicenza, Italy, Jan 2019 - Sep 2019

## Awards and fundings

**Member of the ERC grant**: *sociogeNEsis of criMinal nEtworks: reconStruction, dIscovery and diSruption (NEMESIS)*, Principal investigator: Daniele Durante — 2025 – Present

**Member of the PRIN grant**: *Causes of deAth dependence stRuctures and the cOmpositioNal effecT on ovErall mortality (CARONTE)*, Principal investigator: Daniele Durante — 2024 – 2025

**Member of the PNRR grant**: *Statistical methods and models for the integration of multiomic data*, Principal investigator: Davide Risso — 2023 – 2024

**Member of the PRIN grant**: *Complex Graphical Models for Biological Networks*, Principal investigator: Alberto Roverato — 2023 – 2026

**Merit-based Ph.D. fellowship**, Department of Statistical Sciences, University of Padova — Padova, Italy, 2019 – 2023

**ISBA travel award** at *ISBA 2022 world meeting*. — Montreal, Canada, Jun 2019

**Best Report Prize** at *Stats Under the Stars 3 (SuS3)*. — Florence, Italy, Jun 2019

## Skills and technologies

**Languages:** Italian (native), English (good)

**Programming:** R (advanced), Python (advanced), Julia (advanced), C++ (advanced), Matlab (basic)

**Database:** MySQL (basic)

**Markup:** LaTeX (advanced)

## Publications

### Published articles

Anceschi N., **Castiglione C.**, Rigon T., Zanella G., Durante D. (2026+).
Optimal and computationally tractable lower bounds for logistic log-likelihoods.
*Biometrika* (Accepted, in press). arxiv.org/abs/2410.10309

Segers A., **Castiglione C.**, Vanderaa C., Martens L., Risso D., Clement L. (2026).
omicsGMF: a multi-tool for dimensionality reduction, batch correction and imputation applied to bulk- and single cell proteomics data.
*Nature Communications* (Accepted, in press). (bioRxiv)

**Castiglione, C.**, Segers, A., Clement, L. and Risso, D. (2026).
Stochastic gradient descent estimation of generalized matrix factorization models with application to single-cell RNA sequencing data.
*Biostatistics*, 27(1): kxag010. (Journal)

Di Battista I., De Sanctis M.F., Arnone E., **Castiglione C.**, Palummo A., Sangalli L.M. (2025).
A semiparametric space-time quantile regression model.
*Journal of Nonparametric Statistics*, 38(1): 279–310. (Journal)

**Castiglione, C.**, Bernardi, M. (2025).
Non-conjugate variational Bayes for pseudo-likelihood mixed effect models.
*Journal of Computational and Graphical Statistics*, 35(1): 304–314. (Journal)

De Sanctis M.F., Di Battista I., Arnone E., **Castiglione C.**, Palummo A., Bernardi M., Ieva F., Sangalli L.M. (2025).
Exploring nitrogen dioxide spatial concentration via physics-informed multiple quantile regression.
*Environmental and Ecological Statistics*, 32: 855–892. (Journal)

**Castiglione, C.**, Arnone, E., Bernardi, M., Farcomeni, A., Sangalli, L.M. (2024).
PDE-regularised spatial quantile regression.
*Journal of Multivariate Analysis*, 205: 105381. (Journal)

## Manuscripts

Romanò G., **Castiglione C.**, Durante D. (2026+).
Dependent stochastic block models for age-indexed sequences of directed causes-of-death networks.
arxiv.org/abs/2510.01806 (Under review)

## Conference proceedings and contributed discussions

**Castiglione, C**., Romanò, G. (2025).
Age-dependent analysis of mortality patterns in Italy: a network perspective via dynamic stochastic block models.
*Statistics for Innovation I, SIS 2025, Short Papers, Plenary, Specialized, and Solicited Sessions*, pp. 271–276.
ISBN: 978-3-031-96735-1 (Book)

De Sanctis, M.F., Di Battista, I., Arnone, E., **Castiglione, C.**, Bernardi, M., Palummo, A., Sangalli, L.M. (2025).
Estimating multiple quantile surfaces: A penalized functional approach.
*New Trends in Functional Statistics and Related Fields, Proceedings of IWFOS 2025*, pp. 151–158.
ISBN: 978-3-031-92382-1 (Book)

De Sanctis, M.F., Di Battista, I., Arnone, E., **Castiglione, C.**, Bernardi, M., Palummo, A., Sangalli, L.M. (2024).
Penalised spatial quantile regression: application to air quality data.
*Book of Short Papers 2024, Proceedings of the 53rd Scientific Meeting of the Italian Statistical Society*, pp. 532–537.
ISBN: 978-3-031-64431-3 (Book)

**Castiglione, C.**, Arnone, E., Bernardi, M., Farcomeni, A., Sangalli, L. M. (2023).
Penalized quantile regression for spatially distributed data.
*Book of Short Papers GRASPA 2023, Proceedings of the GRASPA 2023 Conference*, pp. 124–129.
ISBN: 979-12-210-3389-2 (Book)

**Castiglione, C.**, Bernardi, M. (2022).
Probabilistic load forecasting via dynamic quantile regression.
*Book of Short Papers IWSM 2022, Proceedings of the 36th International Workshop on Statistical Modelling*, pp. 400–405.
ISBN: 978-88-5511-309-0 (Book)

**Castiglione, C.**, Bernardi, M. (2022).
Sparse signal extraction via variational SVM.
*Book of Short Papers SIS 2022, Proceedings of the 51th Scientific Meeting of the Italian Statistical Society*, pp. 864–870.
ISBN: 9788891932310 (Book)

Sottosanti, A., Risso, D., **Castiglione, C.** (2022).
Contributed discussion: "Bayesian nonstationary and nonparametric covariance estimation for large spatial data" by Kidd B. and Katzfuss M. *Bayesian Analysis*, 17(1): 337–339. (Book)

**Castiglione, C.**, Bernardi, M. (2021).
Semiparametric variational inference for Bayesian quantile regression.
*Book of Short Papers SIS 2021, Proceedings of the 50th Scientific Meeting of the Italian Statistical Society*, pp. 683–688.
ISBN: 9788891927361 (Book)

### Ongoing projects

**Castiglione C.**, Maestrini L., Bernardi M. (2026+).
On frequentist variational inference for generalized additive models.
*Working paper*.

Bianco N., **Castiglione C.** (2026+).
Improving Bayesian semi-parametric regression via increasing shrinkage priors.
*Working paper*.

## Conference presentations

Romanò, G., **Castiglione, C.**, Durante, D. (2025) – Invited presentation.
Dependent stochastic block models for age-indexed sequences of directed causes-of-death networks.
*The Sixth Meeting of the Multi-Cause Network*, Barcelona, Spain, 16–17 October, 2025.

Romanò, G., **Castiglione, C.**, Durante, D. (2025) – Invited presentation.
Dependent stochastic block models for age-indexed sequences of directed causes-of-death networks.
*Climbing Mortality Models II [final workshop of CARONTE]*, Misurina, Italy, 27–29 August, 2025.

**Castiglione, C**., Romanò, G. (2025) – Invited presentation.
Age-Dependent Analysis of Mortality Patterns in Italy: A Network Perspective via Dynamic Stochastic Block Models.
*SIS 2025. Statistics for Innovation*, Genoa, Italy, 16–18 June, 2025.

**Castiglione, C.**, Romanò, G., Durante, D. (2024) – Invited presentation.
Dynamic stochastic block models with application to causes of death networks.
*18th International Joint Conference CFE-CMStatistics 2024*, London, UK, 14–16 December.

**Castiglione, C.**, Bianco, N. (2024) – Poster presentation.
Improving Bayesian semiparametric regression via increasing shrinkage priors.
*2024 World Meeting of the International Society for Bayesian Analysis (ISBA 2024)*, Venice, Italy, 1–7 July.

**Castiglione, C.**, Arnone, E., Bernardi, M., Farcomeni, A., Sangalli, L. M. (2024) – Invited presentation.
A flexible framework for spatial quantile regression via PDE regularization.
*International Symposium on Nonparametric Statistics (ISNPS 2024)*, Braga, Portugal, 25–29 July.

**Castiglione, C.**, Bianco, N. (2023) – Poster presentation.
Increasing shrinkage in Bayesian nonparametric regression for differential expression analysis.
*2023 IMS International Conference on Statistics and Data Science (ICSDS 2023)*, Lisbon, Portugal, 11–14 November.

**Castiglione, C.**, Arnone, E., Bernardi, M., Farcomeni, A., Sangalli, L. M. (2023) – Poster presentation.
Penalized quantile regression for spatially distributed data.
*Biennial Conference of the Italian Research Group for Environmental Statistics (GRASPA 2023)*, Palermo, Italy, 10–11 July.

**Castiglione, C.**, Bernardi, M. (2023) – Poster presentation.
Approximate belief updating via semiparametric variational Bayes. (poster presentation)
*Greek Stochastics $\nu'$, Contemporary Bayesian Inference*, Naxos, Greece, 7–10 July.

**Castiglione, C.** (2022) – Poster presentation.
Approximate belief updating via semiparametric variational Bayes.
*Statistical Methods and Models for Complex Data 2022*, Padova, Italy, 21–21 September.

**Castiglione, C.**, Bernardi, M. (2022) – Invited presentation.
Approximate general Bayesian inference via semiparametric variational Bayes.
*24th Conference on Computational Statistics (COMPSTAT 2022)*, Bologna, Italy, 23–26 August.

**Castiglione, C.**, Bernardi, M. (2022) – Poster presentation.
Probabilistic load forecasting via dynamic quantile regression.
*36th International Workshop on Statistical Modelling (IWSM 2022)*, Trieste, Italy, 18–22 July.

**Castiglione, C.**, Bernardi, M. (2022) – Contributed presentation.
Approximate general Bayesian inference via semiparametric variational Bayes.
*2022 World Meeting of the International Society for Bayesian Analysis (ISBA 2022)*, Montreal, Canada, 26 June – 1 July.

**Castiglione, C.**, Bernardi, M. (2022) – Contributed presentation.
Sparse signal extraction via Variational SVM.
*51th Scientific Meeting of the Italian Statistical Society (SIS 2022)*, Caserta, Italy, 22–24 June.

**Castiglione, C.** (2021) – Contributed presentation.
Approximate variational inference based on data augmentation methods.
*14th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2021)*, London, UK, 18–20 December.

**Castiglione, C.**, Bernardi, M. (2021) – Poster presentation.
Variational inference for non-crossing quantile regression.
*2021 World Meeting of the International Society for Bayesian Analysis (ISBA 2021)*, Online, 28 June – 02 July.

**Castiglione, C.**, Bernardi, M. (2022) – Contributed presentation.
Semiparametric variational inference for Bayesian quantile regression.
*50th Scientific Meeting of the Italian Statistical Society (SIS 2021)*, Cagliari, Italy, 22–24 June.

## Software

**sgdGMF:** An `R/C++` package for the estimation of high-dimensional generalized matrix factorization models via adaptive stochastic gradient descent. — CRAN package, github/repo

**BayesGLMM:** A `Julia` package for the estimation of Bayesian generalized linear mixed effect models (GLMM) via variational approximations and non-conjugate variations message passing. — github/repo

## Teaching

**Instructor**, 18 hours — Milan, Italy
Bocconi University — Feb 2026 - Jul 2026
Course: *Quantitative Methods for Social Sciences (Module II - Data Analytics)*,
Bachelor in International Politics and Government

**Teaching assistant**, 25 hours — Milan, Italy
Bocconi University — Feb 2026 - Jul 2026
Course: *Quantitative Methods for Social Sciences (Module II - Data Analytics)*,
Bachelor in International Politics and Government

**Instructor**, 6 hours — Milan, Italy
Bocconi University — Feb 2026 - Jul 2026
Course: *Machine Learning (Module I - Introduction)*,
Bachelor in International Politics and Government

**Teaching assistant**, 25 hours — Milan, Italy
Bocconi University — Feb 2026 - Jul 2026
Course: *Machine Learning (Module I - Introduction)*,
Bachelor in International Politics and Government

**Instructor**, 16 hours — Milan, Italy
Bocconi University — Feb 2026 - Jul 2026
Course: *Foundations of Data Science*,
Bachelor in CLEAM, CLEF, CLEACC, BESS-CLES, WBB, BIEF, BIEM, BIG, BEMACS, BAI

**Teaching assistant**, 15 hours — Milan, Italy
Bocconi University — Feb 2026 - Jul 2026
Course: *Foundations of Data Science*,
Bachelor in CLEAM, CLEF, CLEACC, BESS-CLES, WBB, BIEF, BIEM, BIG, BEMACS, BAI

**Instructor**, 2 hours — Milan, Italy
Bocconi University — Feb 2025 - Jul 2025
Course: *Quantitative Methods for Social Sciences (Module II - Data Analytics)*,
Bachelor in International Politics and Government

**Teaching assistant**, 14 hours — Milan, Italy
Bocconi University — Feb 2025 - Jul 2025
Course: *Quantitative Methods for Social Sciences (Module II - Data Analytics)*,
Bachelor in International Politics and Government

**Instructor**, 4 hours — Milan, Italy
Bocconi University — Feb 2025 - Jul 2025
Course: *Machine Learning (Module I - Introduction)*,
Bachelor in International Politics and Government

**Teaching assistant**, 20 hours — Milan, Italy
Bocconi University — Feb 2025 - Jul 2025
Course: *Machine Learning (Module I - Introduction)*,
Bachelor in International Politics and Government

**Teaching assistant**, 10 hours — Milan, Italy
Bocconi University — Feb 2025 - Jul 2025
Course: *Foundations of Data Science*,
Bachelor in CLEAM, CLEF, CLEACC, BESS-CLES, WBB, BIEF, BIEM, BIG, BEMACS, BAI

**Instructor**, 14 hours — Padua, Italy
University of Padua, Department of Statistical Sciences — Oct 2024 - Jan 2025
Course: *Multivariate data analysis*, Bachelor in Statistics

**Instructor**, 22 hours — Padua, Italy
University of Padua, Department of Statistical Sciences — Feb 2024 - Jul 2024
Course: *Statistical Models 1*, Bachelor in Statistics

**Instructor**, 14 hours — Padua, Italy
University of Padua, Department of Statistical Sciences — Oct 2023 - Jan 2024
Course: *Multivariate data analysis*, Bachelor in Statistics

**Academic tutor**, 25 hours — Padua, Italy
University of Padua, Department of Statistical Sciences — Sep 2017 - Sep 2018
Course: *Advanced statistics*, Master in Statistics

**Academic tutor**, 25 hours — Padua, Italy
University of Padua, Department of Statistical Sciences — Sep 2017 - Sep 2018
Course: *Calculus 1*, Bachelor in Statistics

## Supervision

**Master thesis**, program in Mathematical Engineering, Politecnico di Milano — 2023
Title: *Penalised quantile spatial regression: simultaneous estimation and spatio-temporal modelling*
Students: Ilenia Di Battista, Marco F. De Sanctis
Advisors: Prof. Laura M. Sangalli, Eleonora Arnone, **Cristian Castiglione**

## Referee service

Journal of Computational and Graphical Statistics • Bernoulli • Bayesian Analysis • Statistics in Medicine • Journal of Statistical Planning and Inference • Statistical Modelling • Statistical Methods & Applications • STAT • Demonstratio Mathematica.